

JINXUAN ZHU

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EDUCATION BACKGROUND

National University of Singapore

Aug. 2025-Jan.2027(expected)

Master of Computing in Artificial Intelligence (MComp AI) GPA:4.5/5

Singapore

Harbin Institute of Technology, Shenzhen

Sep. 2021-Jul. 2025

BEng in Automation

Shenzhen, China

GPA: 89.7/100 (top10%) Exchange: University of Oxford (Aug 2023); City University of Hong Kong (Jan – May 2025)

RESEARCH INTEREST

I am a **full-stack** robotics researcher mainly focusing on **robotic manipulation**, particularly on (1) integrating foundation models into adaptive robotics system (2) representation and model design for data-driven policies (e.g. VLA) (3) robotic hardware design.

SELECTED PUBLICATIONS

- [1] **Jinxuan Zhu***, Zihao Yan*, Yangyu Xiao, Jingxiang Guo, Chenrui Tie, Xinyi Cao, Yuhang Zheng, Lin Shao, “*ShapeForce: Low-Cost Soft Robotic Wrist for Contact-Rich Manipulation*”. Accepted by International Conference on Robotics and Automation (**ICRA2026**) & (**Best Demo Award Finalist, IROS CIM Workshop 2025**) [[Website](#)] [[Arxiv](#)]
- [2] **Jinxuan Zhu***, Chenrui Tie*, Xinyi Cao*, Yuran Wang, Jingxiang Guo, Zixuan Chen, Haonan Chen, Junting Chen, Yangyu Xiao, Ruihai Wu, Lin Shao, “*AdaptPNP: Integrating Prehensile and Non-Prehensile Skills for Adaptive Robotic Manipulation*”. Accepted by International Conference on Robotics and Automation (**ICRA2026**) [[Website](#)] [[Arxiv](#)]
- [3] Hanyi Zhao*, **Jinxuan Zhu***, Zihao Yan*, Yichen Li, Yuhong Deng, and Xueqian Wang, “*Learning Generalizable Language-Conditioned Cloth Manipulation from Long Demonstrations*”. Accepted by International Conference on Intelligent Robots and Systems (**IROS 2025**) [[Website](#)] [[Arxiv](#)] [[Code](#)]
- [4] Chenrui Tie*, Shengxiang Sun*, **Jinxuan Zhu**, Yiwei Liu, Jingxiang Guo, Yue Hu, Haonan Chen, Junting Chen, Ruihai Wu, Lin Shao, “*Manual2Skill: Learning to Read Manuals and Acquire Robotic Skills for Furniture Assembly Using Vision-Language Models*”. Accepted by Robotics: Science and Systems (**RSS 2025**) [[Website](#)] [[Arxiv](#)] [[Code](#)]
- [5] Chenrui Tie*, Shengxiang Sun*, Yudi Lin, Yanbo Wang, Zhongrui Li, Zhouhan Zhong, **Jinxuan Zhu**, Yiman Pang, Haonan Chen, Junting Chen, Ruihai Wu, Lin Shao. “*Manual2Skill++: Connector-Aware General Robotic Assembly from Instruction Manuals via Vision-Language Models*”. Accepted by International Conference on Robotics and Automation (**ICRA2026**) [[Website](#)] [[Arxiv](#)]

RESEARCH EXPERIENCE

AdaComp Lab & Sharpa (Topic: Whole-Body Dexterous Manipulation)

Dec. 2025 - Present

MComp Thesis (Supervisor: Prof. David Hsu, National University of Singapore)

Singapore

- We introduce OmniCap, a portable system for collecting teleoperation, DAgger correction, and human demonstration data for whole-body dexterous manipulation, and study how these data modalities improve learning efficiency on wheeled humanoid robots.

LinS Lab (Topic: Robotic Furniture Assembly)

Dec. 2024 – Dec.2025

Research Assistant (Supervisor: Prof. Lin Shao, National University of Singapore)

Singapore

- Targeting fully automatic furniture assembly, we explores multiple aspects of furniture assembly. **Manual2Skill (P.4, RSS’25)** and **Manual2Skill++ (P.5, ICRA’2026)** focus on extracting task-relevant information from manuals to guide planning and provide target poses for assembly. **ShapeForce (P.1, ICRA’2026)** aims to develop a low-cost, plug-and-play force sensor for enabling contact-rich tasks. **AdaptPNP (P.2, ICRA’2026)** integrates prehensile and non-prehensile skills to achieve adaptive manipulation.

AI&Robotics Lab (Topic: Generalizable Cloth Manipulation)

Jul. 2024 – Mar. 2025

Research Assistant (Supervisor: Prof. Xueqian Wang, Tsinghua University)

Shenzhen, China

- Developed a method leveraging Large Language Models (LLM) to decompose benchmark and discover basic skills, and subsequently recompose them to multi-step unseen tasks. (**P.3 IROS’25**)

INTERNSHIP

Samsung Research*Research Intern*

Jun. 2026– Present

- Research in Vision-Language-Action Model (VLA) and World-Action-Model (WAM) for robotics manipulation.

Sharpa Robotics*Robot Manipulation Algorithm Intern*

Dec. 2025– Jun.2026

- Contributed to the Blackjack and pinwheel assembly demos using VLTA model [CraftNet](#). Demos can be found [here](#).

RoboScience*Robot Algorithm Intern*

Mar. 2025– Jul.2025

- Developed a fully autonomous IKEA furniture assembly robot system and assembled an IKEA chair. A demo can be found [here](#).

Dobot Robotics*Robot Imitation Learning Algorithm Intern*

Jul. 2024– Nov.2024

- Deployed ACT, DP policies in dual-arm manipulation platform for long-horizon dish washing. A demo can be found [here](#).

MISCELLANEOUS

Professional Skills: Python, C++, C, Linux, IsaacSim, Solidworks, Blender, ROS, Arduino, STM32.**Language:** Chinese (native), English (IELTS 7.5 (6.5))**Academic Service:** Reviewer for ICRA IROS